

Introduction to Databases & Data Engineering

1. Introduction to Databases

A database is an organized collection of data that is stored electronically. Databases are designed to store, retrieve, update, and manage data efficiently. They are used in almost every system today, such as banking systems, school systems, websites, mobile apps, and business systems.

2. What is Data Engineering?

Data Engineering is the field that focuses on collecting, storing, transforming, and making data available for analysis. Data Engineers build pipelines that move data from different sources into databases or data warehouses so that analysts and data scientists can use the data.

3. What is Structured Data?

Structured data is data that is organized in a clear format, usually in rows and columns. It follows a fixed schema and is easy to store in databases. Examples include tables in SQL databases such as student records, employee lists, and sales data.

4. What is Unstructured Data?

Unstructured data does not follow a fixed format or structure. It includes data such as text documents, images, videos, audio files, emails, and social media posts. This type of data is harder to store and analyze using traditional databases.

5. What is Semi-Structured Data?

Semi-structured data falls between structured and unstructured data. It does not follow a strict table format but still contains tags or keys to organize information. Examples include JSON files, XML files, and CSV files.

6. What is SSMS and How Does it Work?

SQL Server Management Studio (SSMS) is a tool used to manage Microsoft SQL Server databases. It allows users to write SQL queries, create databases and tables, insert data, and run reports. SSMS provides a graphical interface to make working with databases easier.

7. Visual Studio Code (VS Code)

Visual Studio Code is a lightweight code editor used for writing and editing code. It supports many programming languages such as SQL, Python, JavaScript, and more. VS Code is commonly used by data engineers to write scripts, manage projects, and connect to databases.

8. Why Databases are Important

Databases are important because they allow organizations to store large amounts of data securely and efficiently. They help ensure data accuracy, reduce duplication, and allow multiple users to access data at the same time.